

Cherry Blossom – Two Proportions Part 2

MATH 213

Group 1

Paste your dotplot and summary statistics from the R activity “2 proportions part 2” here, and answer the questions on the next slide.

1. What does each dot in the dotplot for the sampling distribution represent?
2. Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?
3. Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape?

Hopefully, on the previous slide, you said that you would expect the shape of the sampling distribution to be symmetrical, bell-shaped. More specifically, we said that sampling distributions tend to be **normal**.

4. If data is distributed according to a **normal distribution**, as you will recall, then 95% of the data should fall between **how many** standard deviations of the mean?

5. Does our graph have this feature? Check this and explain. (You will need to remember the special name given to the standard deviation of a sampling distribution.)

a. Special name for std deviation of a sampling distribution:

a. Check to see if our sampling distribution is normal:

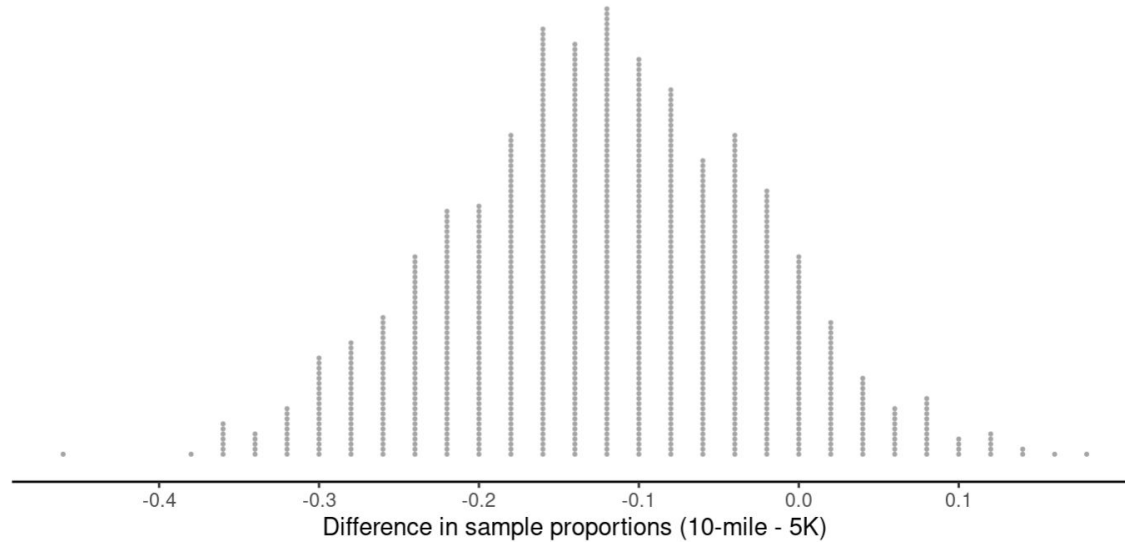
Group 2

Paste your dotplot and summary statistics from the R activity “2 proportions part 2” here, and answer the questions on the next slide.



Sampling distribution for the difference in proportions

Proportion of 'F' in 10-mile sample - proportion of 'F' in 5K sample



1. What does each dot in the dotplot for the sampling distribution represent?

Each dot contains the difference in proportion of male to female runners between the 10m and 5k events from random samples of 50 people

2. Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

We would expect there to be no difference between proportions, meaning the center of his distribution would be at 0.0. This is true from our graph

3. Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape?

We would expect this sampling distribution to be bell shaped, which is true.

Hopefully, on the previous slide, you said that you would expect the shape of the sampling distribution to be symmetrical, bell-shaped. More specifically, we said that sampling distributions tend to be **normal**.

4. If data is distributed according to a **normal distribution**, as you will recall, then 95% of the data should fall between **how many** standard deviations of the mean?

95% of the data should fall between 2 Standard deviations of the mean

5. Does our graph have this feature? Check this and explain. (You will need to remember the special name given to the standard deviation of a sampling distribution.)

a. Special name for std deviation of a sampling distribution:
The standard error

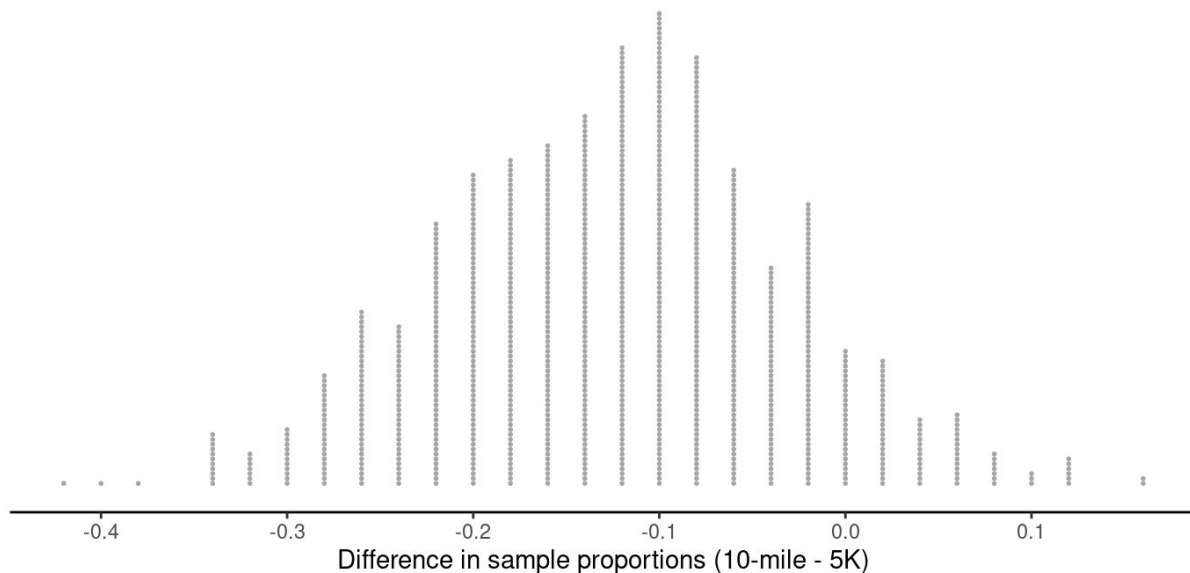
a. Check to see if our sampling distribution is normal:

In ht simulations the results fell between

Group 3

Paste your dotplot and summary statistics from the R activity “2 proportions part 2” here, and answer the questions on the next slide.

Sampling distribution for the difference in proportions
Proportion of 'F' in 10-mile sample - proportion of 'F' in 5K sample



Q025: -0.3

Q1: -0.18

Median: -0.12

Q3: -0.06

Q975: 0.06

1. What does each dot in the dotplot for the sampling distribution represent?

Each plot represents a single simulation, which are the differences in the sample proportion of female numbers between 10-mile sample and 5k sample from that simulated pair of random samples.

2. Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

You would expect it to be near the true difference in population proportions of female runners. The median is around -0.12, which where the center is.

3. Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape?

You would expect it to be roughly symmetric and bell-shaped. The dotplot confirms the suspicion and is symmetric and bell-shaped.

Hopefully, on the previous slide, you said that you would expect the shape of the sampling distribution to be symmetrical, bell-shaped. More specifically, we said that sampling distributions tend to be **normal**.

4. If data is distributed according to a **normal distribution**, as you will recall, then 95% of the data should fall between **how many** standard deviations of the mean?

95% of the data should fall between 2 standard deviations of the mean.

5. Does our graph have this feature? Check this and explain. (You will need to remember the special name given to the standard deviation of a sampling distribution.)

a. Special name for std deviation of a sampling distribution:
Standard error

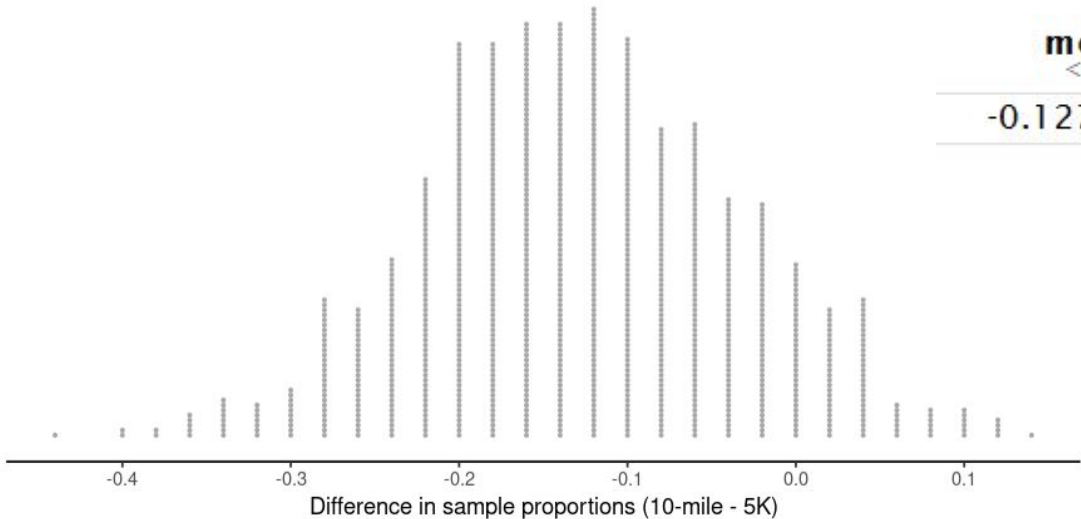
a. Check to see if our sampling distribution is normal:
In my simulation, 95% of the differences in the sample proportion fall between -0.3 and 0.06, which is within the two standard errors of the mean. Showing this sampling distribution is normal.

Group 4

Paste your dotplot and summary statistics from the R activity “2 proportions part 2” here, and answer the questions on the next slide.

quantile_025<dbl>	Q1<dbl>	median<dbl>	Q3<dbl>	quantile_975<dbl>
-0.3005	-0.2	-0.14	-0.06	0.04

Sampling distribution for the difference in proportions
Proportion of 'F' in 10-mile sample - proportion of 'F' in 5K sample



mean<dbl>	se<dbl>
-0.12752	0.09363449

1. What does each dot in the dotplot for the sampling distribution represent?

A difference in the gender proportions (proportion that are female) between the simulated 10mi and 5k races.

2. Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

The center should be roughly at the same value as the difference in gender proportions (p for the difference of women participating both races) for the real world race data.

3. Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape?

A normal standard bell curve

Hopefully, on the previous slide, you said that you would expect the shape of the sampling distribution to be symmetrical, bell-shaped. More specifically, we said that sampling distributions tend to be **normal**.

4. If data is distributed according to a **normal distribution**, as you will recall, then 95% of the data should fall between **how many** standard deviations of the mean?

±2

5. Does our graph have this feature? Check this and explain. (You will need to remember the special name given to the standard deviation of a sampling distribution.)

a. Special name for std deviation of a sampling distribution: Standard Error (SE)

a. Check to see if our sampling distribution is normal:

$$(-0.12) - (0.094*2) = (-0.12) - 0.188 = -0.308$$

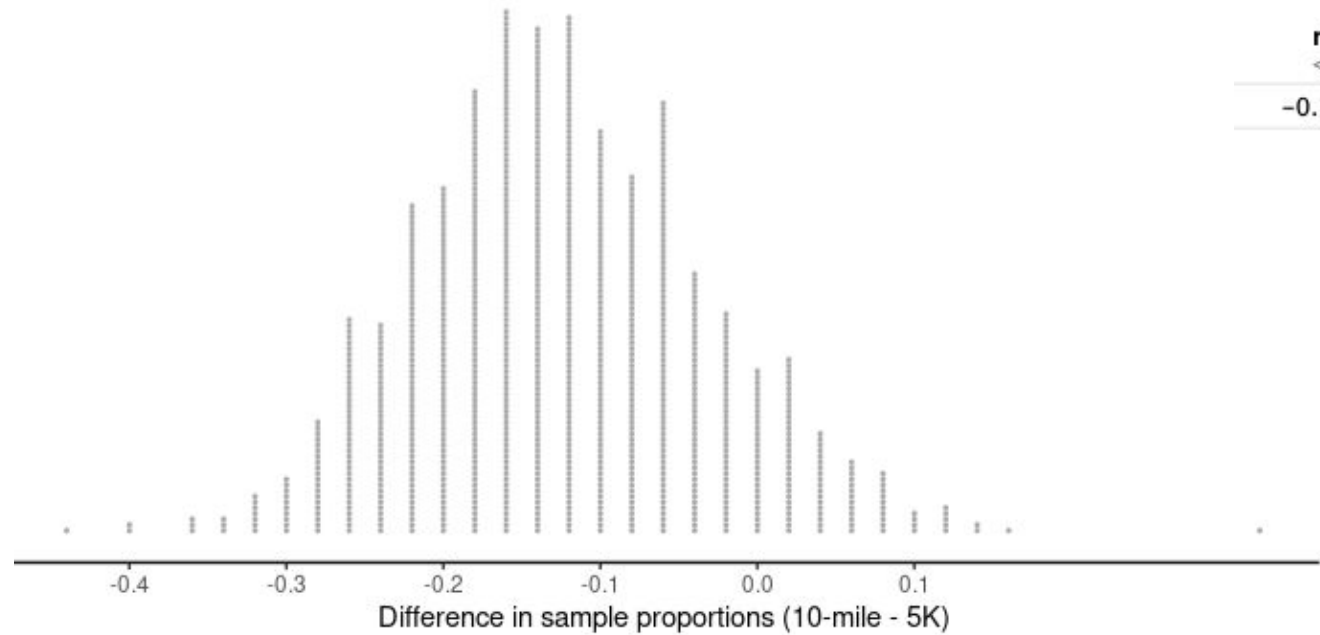
$$(-0.12) + (0.094*2) = (-0.12) + 0.188 = 0.068 \text{ (appears to not be entirely normal w/ a slight skew)}$$

Group 5

Paste your dotplot and summary statistics from the R activity “2 proportions part 2” here, and answer the questions on the next slide.

quantile_025 <dbl>	Q1 <dbl>	median <dbl>	Q3 <dbl>	quantile_975 <dbl>
-0.3	-0.18	-0.12	-0.06	0.06

mean <dbl>	se <dbl>
-0.1246	0.09335714



1. What does each dot in the dotplot for the sampling distribution represent?

The difference in proportions of a sample.

2. Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

It is where expected, since the center should be where the center is for the population proportion difference.

3. Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape?

Bell shaped, however it is slightly left skewed.

Hopefully, on the previous slide, you said that you would expect the shape of the sampling distribution to be symmetrical, bell-shaped. More specifically, we said that sampling distributions tend to be **normal**.

4. If data is distributed according to a **normal distribution**, as you will recall, then 95% of the data should fall between **how many** standard deviations of the mean?

2 standard deviations

5. Does our graph have this feature? Check this and explain. (You will need to remember the special name given to the standard deviation of a sampling distribution.)

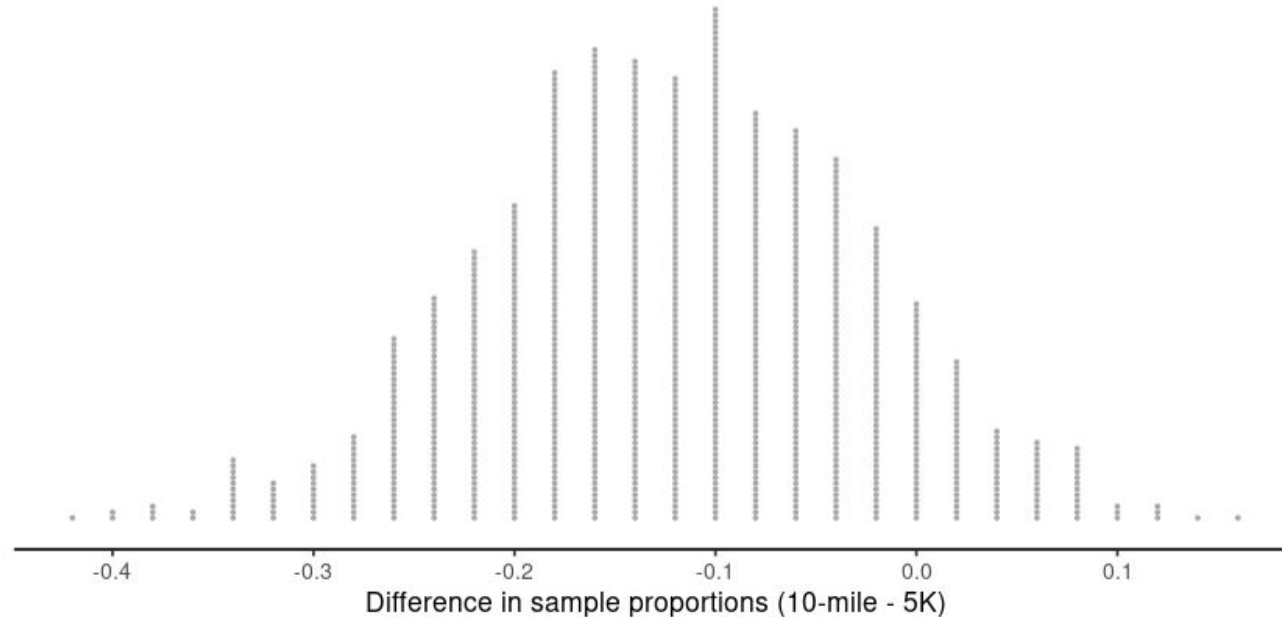
a. Special name for std deviation of a sampling distribution:
Standard error

a. Check to see if our sampling distribution is normal:

Group 6

Paste your dotplot and summary statistics from the R activity “2 proportions part 2” here, and answer the questions on the next slide.

Sampling distribution for the difference in proportions
Proportion of 'F' in 10-mile sample - proportion of 'F' in 5K sample



1. What does each dot in the dotplot for the sampling distribution represent?

Each dot represents the difference in proportions for a sample ($n=50$)

2. Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

The center of the sampling distribution would be at the population difference in proportion, which is -0.1240067 .

So yes, it is where we would expect.

3. Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape?

Bell shaped. The sampling distribution does have a bell shape.

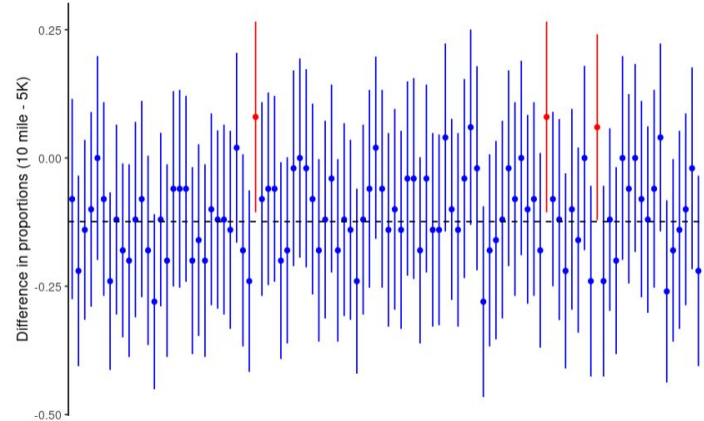
Hopefully, on the previous slide, you said that you would expect the shape of the sampling distribution to be symmetrical, bell-shaped. More specifically, we said that sampling distributions tend to be **normal**.

4. If data is distributed according to a **normal distribution**, as you will recall, then 95% of the data should fall between **how many** standard deviations of the mean? 2

5. Does our graph have this feature? Check this and explain. (You will need to remember the special name given to the standard deviation of a sampling distribution.)

a. Special name for std deviation of a sampling distribution:
Standard error?

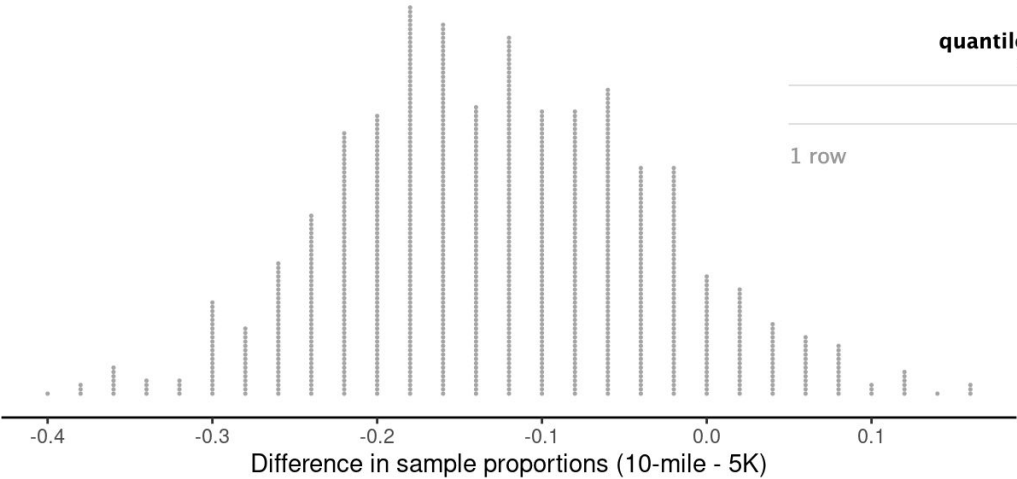
a. Check to see if our sampling distribution is normal:



Group 7

Paste your dotplot and summary statistics from the R activity “2 proportions part 2” here, and answer the questions on the next slide.

Sampling distribution for the difference in proportions
Proportion of 'F' in 10-mile sample - proportion of 'F' in 5K sample



quantile_025<dbl>	Q1<dbl>	median<dbl>	Q3<dbl>	quantile_975<dbl>
-0.3	-0.2	-0.12	-0.06	0.0605

1 row

mean<dbl>	se<dbl>
-0.12634	0.09575267

1 row

1. What does each dot in the dotplot for the sampling distribution represent?

Each dot represents a difference in sample proportions of a random sample of 50

2. Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

-0.12. Yes it is

3. Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape?

It should have a normal bell curve, which it mostly does

Hopefully, on the previous slide, you said that you would expect the shape of the sampling distribution to be symmetrical, bell-shaped. More specifically, we said that sampling distributions tend to be **normal**.

4. If data is distributed according to a **normal distribution**, as you will recall, then 95% of the data should fall between **how many** standard deviations of the mean?

+2SE of the mean

5. Does our graph have this feature? Check this and explain. (You will need to remember the special name given to the standard deviation of a sampling distribution.)

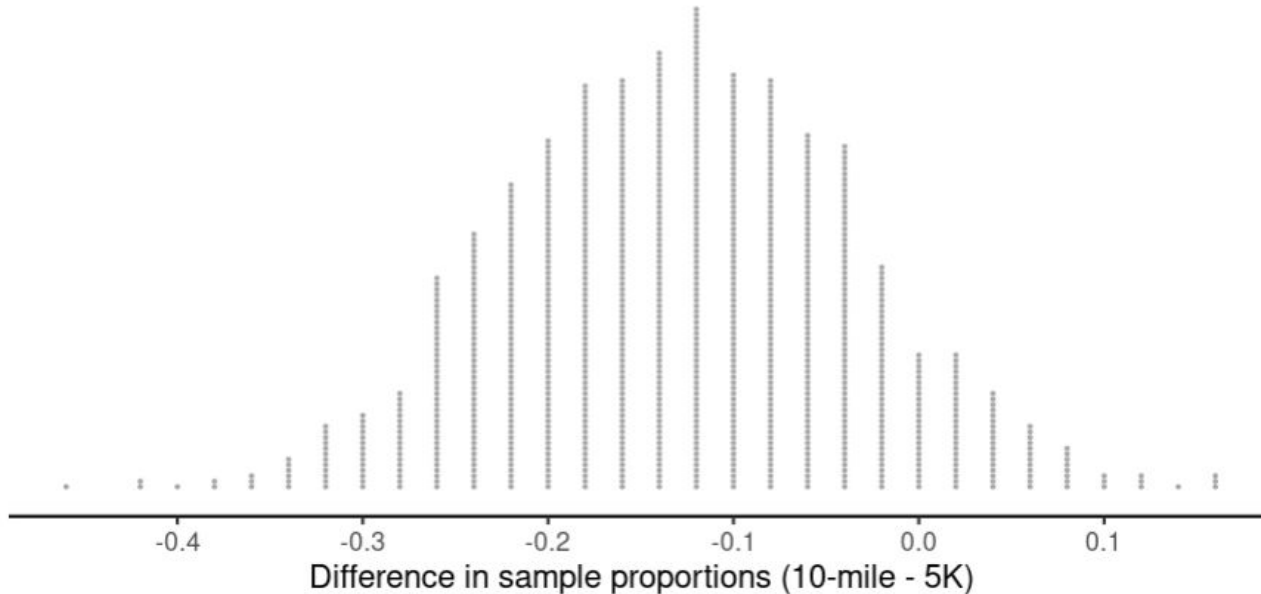
a. Special name for std deviation of a sampling distribution:
Standard error

a. Check to see if our sampling distribution is normal:

Group 8

Paste your dotplot and summary statistics from the R activity “2 proportions part 2” here, and answer the questions on the next slide.

Proportion of 'F' in 10-mile sample - proportion of 'F' in 5K sample



1. What does each dot in the dotplot for the sampling distribution represent?

Each dot is a difference in proportions of sex from many different samples that take different amount of males and females for either a 5k run or 10k

2. Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

I would expect the center to be 0, it isn't where I expected the center is a little less than -1.

3. Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape?

You could expect the shape to be bell shaped with most of our sampling distributions have been, with center always being at the center.

Hopefully, on the previous slide, you said that you would expect the shape of the sampling distribution to be symmetrical, bell-shaped. More specifically, we said that sampling distributions tend to be **normal**.

4. If data is distributed according to a **normal distribution**, as you will recall, then 95% of the data should fall between **how many** standard deviations of the mean?

2 standard deviations

5. Does our graph have this feature? Check this and explain. (You will need to remember the special name given to the standard deviation of a sampling distribution.)

a. Special name for std deviation of a sampling distribution:
Standard error

a. Check to see if our sampling distribution is normal:

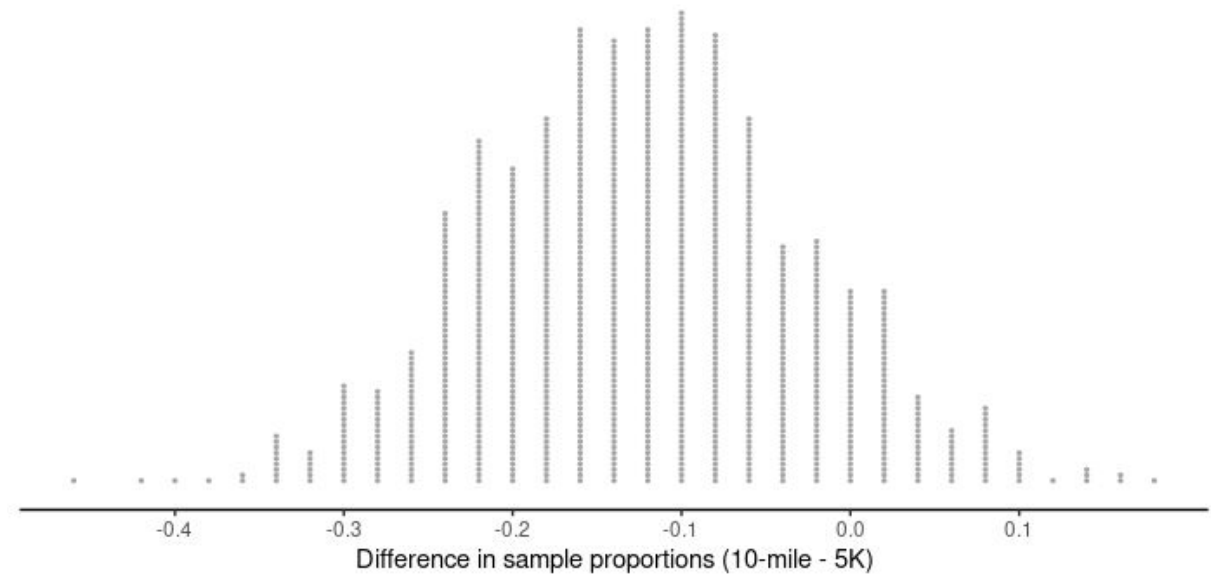
$$-0.13 + 2(0.093) = 0.056$$

$$-0.13 - 2(0.093) = -0.0316$$

Group 9

Paste your dotplot and summary statistics from the R activity “2 proportions part 2” here, and answer the questions on the next slide.

Sampling distribution for the difference in proportions
Proportion of 'F' in 10-mile sample - proportion of 'F' in 5K sample



quantile_025	Q1	median	Q3	quantile_975
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
-0.3	-0.18	-0.12	-0.06	0.08

mean	se
<dbl>	<dbl>
-0.12344	0.09548447

1. What does each dot in the dotplot for the sampling distribution represent?

The difference in proportion from one set of samples for the 10 Mile and 5K races

2. Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

-0.12. Yes

3. Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape?

Bell-shaped/normal. Yes.

Hopefully, on the previous slide, you said that you would expect the shape of the sampling distribution to be symmetrical, bell-shaped. More specifically, we said that sampling distributions tend to be **normal**.

4. If data is distributed according to a **normal distribution**, as you will recall, then 95% of the data should fall between **how many** standard deviations of the mean?

~2

5. Does our graph have this feature? Check this and explain. (You will need to remember the special name given to the standard deviation of a sampling distribution.)

a. Special name for std deviation of a sampling distribution:

Standard Error

b. Check to see if our sampling distribution is normal:

$$-0.12 - 0.09548447 * 2 = -0.31096894 \mid q_{025} = -0.3$$

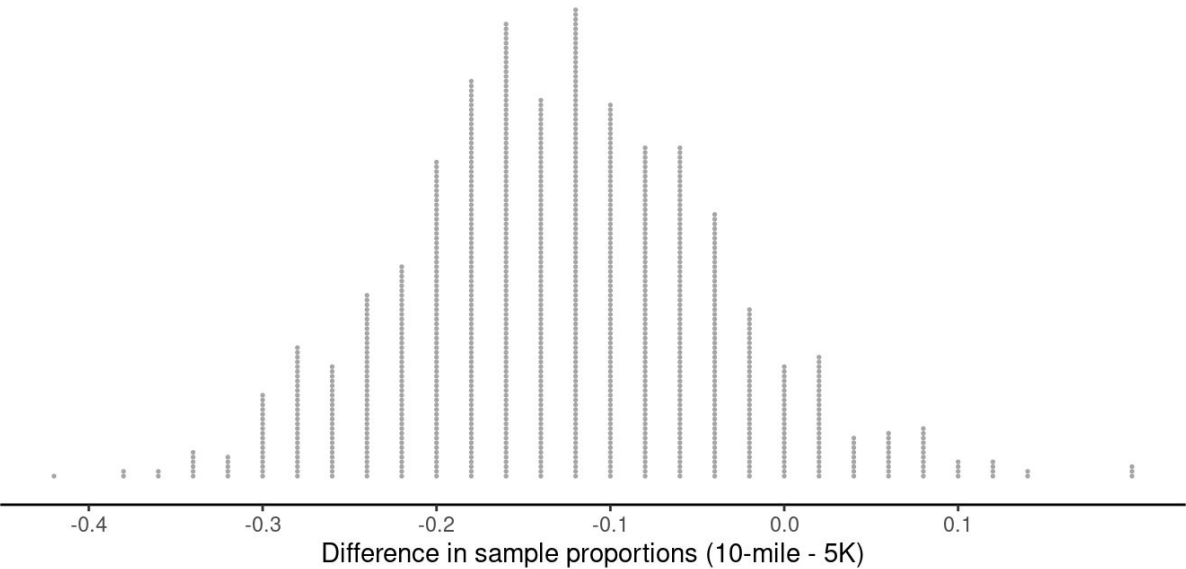
$$-0.12 + 0.09548447 * 2 = 0.07096894 \mid q_{975} = 0.08$$

Group 10

Paste your dotplot and summary statistics from the R activity “2 proportions part 2” here, and answer the questions on the next slide.

Sampling distribution for the difference in proportions

Proportion of 'F' in 10-mile sample - proportion of 'F' in 5K sample



quantile_025 <dbl>	Q1 <dbl>	median <dbl>	Q3 <dbl>	quantile_975 <dbl>
-0.3	-0.18	-0.12	-0.06	0.06
1 row				

bble: 1 × 2

mean <dbl>	se <dbl>
-0.12778	0.09114482
W	

1. What does each dot in the dotplot for the sampling distribution represent?

The difference in proportions of females between the 10mile and the 5k races per a sample

2. Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

0, and I'd say yeah, that's about where it is.

3. Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape?

Bell curve and yeah

Hopefully, on the previous slide, you said that you would expect the shape of the sampling distribution to be symmetrical, bell-shaped. More specifically, we said that sampling distributions tend to be **normal**.

4. If data is distributed according to a **normal distribution**, as you will recall, then 95% of the data should fall between **how many** standard deviations of the mean?

2

5. Does our graph have this feature? Check this and explain. (You will need to remember the special name given to the standard deviation of a sampling distribution.)

a. Special name for std deviation of a sampling distribution: standard error

a. Check to see if our sampling distribution is normal: 0.09114482, and yeah, I think it's normal